AI Exposure or Occupational Composition? Young-Worker Employment Comparisons in the CPS

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Abstract. Recent evidence links generative-AI exposure to weaker employment among workers in their early twenties. I reconstruct that comparison in public monthly Current Population Survey data and ask which occupational contrasts support it. The highest-versus-lowest GPT-4-exposure quintile estimate is -0.132 log points. Allowing broad occupation families their own young-relative monthly paths yields -0.022, and the paired movement is statistically distinguishable from zero. Yet direct tail support spans only 29 occupations and is too imprecise to resolve a within-family gradient. Conditioning on a preperiod computer-use measure makes the association more negative rather than attenuating it, while broader industry conditioning moves it toward zero. Unrestricted quarterly dynamics reject parallel evolution before the public release of ChatGPT, and linked CPS estimates do not distinguish employment exit, reported occupation change, or allocation across destinations conditional on entry. Together, these results show that neither attenuation nor coefficient survival supplies a causal technology decomposition. The public-CPS pattern is therefore a descriptive comparison across broad occupational structure, not an identified causal effect of artificial intelligence or a replication of employer hiring results.

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1 Introduction

Recent evidence associates generative-AI exposure with weaker employment growth among workers in their early twenties. That pattern may signal changing access to entry-level work, but it is difficult to interpret: occupational exposure is constructed rather than observed, high- and low-exposure jobs lie in different parts of the economy, and the period around ChatGPT's release includes pandemic recovery, monetary tightening, return-to-office changes, and survey disruptions.

This paper asks which occupational comparisons support the public Current Population Survey (CPS) pattern. I rebuild monthly employment stocks from January 2017 through July 2026, compare ages 22–25 with ages 26–65 within occupation, and classify occupations with the GPT-4 beta score of Eloundou et al. (2024). I then add broad-family paths, examine direct within-family support, condition on preperiod computer use and other characteristics, estimate unrestricted dynamics, and study linked-CPS flow margins. Exposure measures potential susceptibility, not adoption; January 2023 is chronology, not an instrument.

The reconstructed highest-versus-lowest exposure-quintile coefficient is -0.132 log points, with 95-percent occupation wild-score interval $[-0.221, -0.044]$. It is a change in the young-to-older employment-stock ratio, not an individual employment probability or employer hiring rate. Once each two-digit occupation family receives its own young-relative monthly path, the coefficient is -0.022 ($[-0.161, 0.117]$). The paired movement is 0.110 ($[0.011, 0.210]$), so the conditional estimands are distinguishable even though the residual coefficient is not. This is not a causal decomposition: family paths may absorb AI-related and non-AI-related evolution alike.

Direct support is thin. Only four families span both exposure tails; their direct-tail sample contains 29 occupations and 5.03 percent of preperiod stock. Its interval includes large effects of either sign. The remaining conditioned comparison is linked through intermediate quintiles and common-coefficient restrictions, preventing the near-zero point estimate from being read as economic equivalence or absence of an AI effect.

Computerization is not resolved by one control. On common support, adding a pre-2022 O*NET computer-use measure makes the exposure coefficient more negative. Remotability and other occupational characteristics are correlated descriptions that may be confounds, channels, or both. Unrestricted quarterly coefficients also reject joint preperiod equality, while adjacent- and twelve-month CPS flow intervals include zero. The data therefore neither isolate an AI counterfactual nor attribute the stock pattern to exit, switching, or entry allocation.

The contribution is comparison-centered. The paper shows that a sizeable public-CPS association rests mainly on broad occupational structure, quantifies the limited resolution of direct within-family and flow evidence, and records calendar and taxonomy failures that otherwise alter support silently. It narrows rather than overturns the motivating evidence: it does not replicate proprietary payroll outcomes, identify an AI effect, or establish computerization or pandemic recovery as the cause.

2 Related Evidence and the Empirical Object

The closest motivating study is the August 2026 revision of Brynjolfsson, Chandar, and Chen (2026). Using ADP payroll records through June 2026, it reports weaker employment growth among ages 22–25 in occupations with high GPT-4 beta exposure and studies employer-match hiring and separation. The CPS cannot reproduce its employer panel, standardized-title mapping, firm controls, or worker–firm flows. Because that study also reports public CPS and ACS evidence, novelty here lies in auditing a fixed national-CPS estimand, not discovering the age pattern.

Administrative and workplace studies observe adoption or tool access in particular settings (Humlum and Vestergaard 2025; Brynjolfsson, Li, and Raymond 2025); occupation scores instead describe tasks a technology might affect. Remote-work evidence supplies another competing interpretation (Emanuel, Harrington, and Pallais 2026). None turns a capability score into observed deployment.

The primary score is Eloundou et al.'s GPT-4 beta: tasks are labeled by whether an LLM, alone or with complementary software, could reduce completion time at constant quality (Eloundou et al. 2024). AIOE maps AI applications to abilities (Felten, Raj, and Seamans 2018, 2021); Webb measures patent–task overlap (Webb 2020); Dingel–Neiman classifies teleworkability (Dingel and Neiman 2020). Computer use, routine content, education, and remotability are therefore overlapping occupational characteristics, not clean alternative treatments. The task literature likewise allows technology to substitute for some tasks, complement others, and reallocate work within occupations (Autor, Levy, and Murnane 2003; Autor and Dorn 2013; Acemoglu and Restrepo 2019).

This distinction changes the burden of interpretation. Agreement across correlated scores can show that a direction is not unique to one coding rule, but cannot validate a latent exposure construct. Conversely, coefficient differences can arise from task definitions, occupational support, or treatment-group membership. The empirical work below therefore matches supports where possible and names population changes where it is not.

I distinguish six layers:

$$C \longrightarrow M \longrightarrow H \longrightarrow \Omega \longrightarrow R \longrightarrow Q,$$

the intended construct, measurement rule, occupational harmonization, retained support, representation, and regression estimand. A vintage mismatch is an implementation error; score repair on fixed support changes measurement; occupation readmission changes population; and a different exposure contrast changes the estimand. Common-draw paired inference tests coefficient movement, while each component interval measures its own resolution. Neither a detected movement nor a nondetection assigns a causal share or establishes equivalence. This focus connects the exercise to specification-sensitivity and generated-covariate inference (Andrews, Gentzkow, and Shapiro 2017; Rambachan and Roth 2023; Battaglia et al. 2024; Christensen and Hansen 2026; Duan and Pelger 2026; Ludwig, Mullainathan, and Rambachan 2026).

3 Data and Exposure Construction

3.1 CPS stocks and calendar

I use IPUMS Basic Monthly CPS data from January 2017 through July 2026. Employed respondents ages 18–65 with positive final weights are aggregated to occupation–month stocks; the regression compares ages 22–25 with ages 26–65. Age is not experience, and older workers need not be untreated.

The corrected source calendar contains 114 of 115 possible months. Five March files in the wide extract were ASEC samples and are replaced by authenticated Basic Monthly files for 2017–2021. October 2025 is absent because no CPS was collected during the funding lapse; it is not interpolated. December 2022 is observed but excluded as the transition month, leaving 113 static months.

The reconstruction scans 9,843,021 source rows and retains 5,322,047 eligible records before routing. Fractional pre-2020 crosswalk descendants are not distinct respondents. The 468-occupation grid contains 52,884 cells; 51,891 enter the likelihood, including 12,965 valid one-sided zeros, while 993 both-age-zero cells contribute nothing. Young cells are sparse: 26.25 percent have zero respondent-equivalent records and 69.83 percent have fewer than five.

TABLE 1. Corrected CPS reconstruction and analysis support

	Count	Definition
Expected calendar months	115	Jan. 2017–July 2026
Observed source months	114	October 2025 absent
Static analysis months	113	December 2022 excluded
Eligible employed source records	5,322,047	ages 18–65, positive final weight
Route-expanded rows	6,188,956	fractional descendants allowed
Primary occupations	468	finite beta and Webb; positive pre-stock
Occupation–month grid	52,884	468 occupations $\times$ 113 months
Positive-total fitted cells	51,891	one-sided zeros retained
Both-age-zero cells	993	no likelihood contribution

Notes: Source records are not independent observations for regression inference. Before 2020, one source record can generate several fractional crosswalk descendants. Counts, routing checks, named support, exclusions, and restricted-input hashes are generated by the corrected-calendar data audit.

3.2 Harmonization and support

Pre-2020 Census-2010 occupations are routed to Census-2018 using versioned official shares. Applying the same shares to young and older records preserves source stock and age ratios by construction, but does not validate age-specific routing. The route reaches 99.546 percent of eligible early stock and conserves routed weight to machine precision.

Primary support requires finite GPT-4 beta and Webb software scores and positive preperiod young and older stock. It contains 468 occupations and 87.19 percent of candidate-universe stock; a named file records all 73 exclusions. A literal SOC-2010 AIOE merge into post-2018 employment data covers only 3.33 percent of computer-and-mathematical employment, versus 97.7 percent after version-aware routing. Fixed-support score repair changes little; occupation readmission changes the population. I treat the literal merge as invalid and do not attribute it to another paper without implementation evidence.

The support file separates missing exposure, missing Webb, and absent young preperiod stock. This matters because a common-support regression estimates a population defined by all required scores, whereas native-support rows can change both the exposure definition and who remains. The appendix's four-step crosswalk table holds support fixed before adding newly reachable occupations, so measurement correction is not conflated with sample composition.

3.3 Exposure groups and production breaks

The primary exposure is Eloundou et al.'s GPT-4 beta share. CPS stock from January 2017 through November 2022 sets normalization and employment-weighted quintile cuts, with ties retained. The quintiles contain 129, 99, 82, 67, and 91 occupations and about one fifth of preperiod stock each. No postperiod stock enters the rebuilt labels.

Employment weighting balances stock rather than occupation counts. Ties are assigned to the lower group at a cut and never split to force exact fifths. Consequently, group membership is a versioned treatment object rather than a percentile transformation that can be recreated from rounded published values.

A historical production treatment instead used the full static panel for weights. On the corrected calendar its coefficient is -0.1346. Rebuilding from the 71-month preperiod changes nine assignments and moves the coefficient by 0.0024 log points ($[-0.0013, 0.0062]$). The revised analysis uses only the rebuilt treatment; the historical row is a checkpoint.

Official documentation also identifies population-control changes in January 2025 and January 2026, modified November 2025 collection and weighting, and later longitudinal-identifier corrections. These establish survey-production breaks, not their causal effect on the coefficient. I report endpoint and shutdown-month sensitivities, preserve elapsed calendar time in lags, and do not apply aggregate population adjustments to age-by-occupation cells without the required counterfactual controls.

4 Estimand and Inference

Let N_{oat} be CPS-weighted employment stock in occupation o, age group a, and month t; Y_a indicate ages 22–25; P_t indicate January 2023 onward; Q_{oq} indicate rebuilt exposure quintile q; and W_o be standardized Webb software exposure. The pooled mean is

$$E[N_{oat} \mid X] = \exp\left[\gamma_{oa} + \delta_{ot} + \lambda_{at} + \sum_{q=2}^{5} \beta_q Q_{oq} Y_a P_t + \theta W_o Y_a P_t\right]. \tag{1}$$

The target β_5 is the postperiod change in the young-to-older stock ratio for Q5 relative to Q1, conditional on Q2–Q4 and Webb. It is not an individual probability, continuous slope, or hiring rate.

With two age groups, conditioning on $T_{ot} = N_{oyt} + N_{oot}$ removes the occupation–month effect and gives

$$\ell_{ot} = N_{oyt} \log p_{ot} + N_{oot} \log(1 - p_{ot}), \qquad \operatorname{logit}(p_{ot}) = \log(\mu_{oyt}/\mu_{oot}). \tag{2}$$

Final weights construct continuous cell stocks once; equation (2) is an estimating criterion, not a literal independent-binomial sampling model.

The central alternative gives each SOC two-digit family its own young-relative calendar-month path; a less saturated companion uses family-specific post shifts. Both change the conditional estimand. Neither the residual coefficient nor its movement from equation (1) is an AI remainder or composition share.

The occupational-conditioning, influence, dynamic, and flow analyses added in this revision were designed after the initial outcome pattern was known. They are post-outcome exploratory analyses, not preregistered confirmatory tests; simultaneous inference does not undo that selection.

For shared observations I reuse bootstrap multipliers and form each draw of $\Delta = \widehat{\beta}^{(2)} - \widehat{\beta}^{(1)}$. A paired interval excluding zero shows statistical distinguishability under that procedure; it does not identify the reason. An interval including zero means nondetection, not economic equivalence. I report $\text{MDE}_{80} \simeq (1.96 + 0.84)SE$ only as a precision diagnostic.

To describe where a target is learned, let $h_{ot} = T_{ot} p_{ot}(1 - p_{ot})$ and let r_{ot} be the target regressor after projection on every nuisance regressor in the fitted-information metric. I report

$$I = \sum_{o,t} h_{ot} r_{ot}^2, \qquad s_o = \frac{\sum_t h_{ot} r_{ot}^2}{I}, \qquad G_{\text{eff}} = \left(\sum_o s_o^2 \right)^{-1}.$$

These are fitted target information and concentration diagnostics, not an effective number of CPS sampling clusters. Family and occupation shares, leave-one-out estimates, and the top-five information share are reported without using influence to select a preferred sample.

The principal 9,999-draw wild-score interval treats occupations as shock clusters and allows arbitrary serial dependence within occupation. It conditions on weighted cells, exposure labels, mapping, support, and specification. Separate sensitivities use 22 broad-family shocks, household/sample-unit full refits under released weights, and elapsed-calendar time-HAC scores. A sparse-cell and family-shock simulation audits finite-sample behavior. These procedures have different stochastic targets; they are not mechanically combined, and none supplies complete CPS design inference or exposure-label uncertainty.

The household exercise keeps every month and fractional route descendant of a positive CPS household identifier together. It still omits unavailable multistage-selection and calibration uncertainty. The time-HAC construction uses elapsed calendar lags, so the missing October 2025 is not treated as adjacency between September and November.

5 Which Occupational Comparisons Support the Pattern?

Table 2 begins with the rebuilt pooled result. The Q5–Q1 coefficient is -0.1321, its occupation-clustered standard error is 0.0452, and the 9,999-draw interval is $[-0.2206, -0.0437]$. Giving every SOC two-digit family its own young-relative calendar-month path yields -0.0217 ($[-0.1607, 0.1173]$). On common draws, the conditioned-minus-pooled movement is 0.1104 ($[0.0107, 0.2102]$), with paired $\text{MDE}_{80} = 0.1454$. Thus the conditional estimands are distinguishable while the residual gradient is imprecise.

The ratio 0.1104/0.1321 is not a causal "share explained." Broad families differ in AI-relevant tasks, computerization, education, work location, and pandemic recovery; family paths absorb all such common evolution. The result says that the pooled estimate relies mainly on comparisons across families, not what would have happened without AI.

The full Q2–Q5 profile reaches the same conclusion. Pooled coefficients are jointly different from zero but do not establish a monotone dose response. After family-month conditioning,

TABLE 2. Pooled and broad-family exposure comparisons

Specification	Coefficient	95% interval	Paired movement	MDE_{80}
Pooled Q5–Q1	−0.1321	$[-0.2206, -0.0437]$	—	0.1266
SOC2 × young × month	−0.0217	$[-0.1607, 0.1173]$	0.1104	0.1998

Notes: The dependent variable is the log young-to-older employment-stock ratio implicit in the grouped-binomial model. The second row changes the conditioning estimand. Intervals use 9,999 occupation wild-score draws. The paired movement is the second-row coefficient minus the first on common draws; its interval is $[0.0107, 0.2102]$ and its MDE_{80} is 0.1454. MDEs are normal-theory precision diagnostics, not rejection thresholds.

the four coefficients are jointly indistinguishable from zero. Simultaneous intervals and paired movements appear in the online appendix.

5.1 Direct support and restrictions

Only SOC27, 29, 31, and 41 contain Q1 and Q5 occupations. Elsewhere the regression links partial segments—for example Q1–Q3 in one family and Q4–Q5 in another—through common quintile coefficients. That is valid regression variation but not repeated side-by-side tail comparisons.

A direct-tail benchmark uses only Q1 and Q5 occupations in the four spanning families. Its 29 occupations account for 5.03 percent of preperiod stock. The family-month estimate is 0.1494 ($[-0.1694, 0.4682]$), so the CPS cannot precisely resolve a direct tail contrast within broad families. This row changes population and support and cannot replace the primary estimand.

A continuous companion instead subtracts each family's preperiod-weighted beta mean and imposes a common within-family slope. Its family-month coefficient is -0.0025 per 0.1084 raw beta units ($[-0.0240, 0.0191]$). It uses more residual variation but trades the tail contrast for linearity and a common slope. Together, the specifications expose the support–functional-form tradeoff.

5.2 Influence and observed paths

The online appendix reports nuisance-adjusted information, effective information counts, and family and occupation deletions. Fast-food and counter workers have substantial signed influence because low-exposure service employment recovered differently after the pandemic, but other occupations push oppositely. Outcome-selected deletion is not a design correction; symmetric trimming and economically defined service exclusions remain diagnostics.

Figure 1 separates young and older stocks for both tails. Because numerator and denominator move, the coefficient is not an individual young worker's loss. The low-exposure reference trajectory is part of the result.

6 Characteristics and Inferential Resolution

On the focused 408-occupation support, beta's preperiod-weighted correlation is 0.794 with O*NET computer use and 0.589 with remotability. A static horse race is therefore descriptive: residual variation may be narrow, and a control may also be an AI channel.

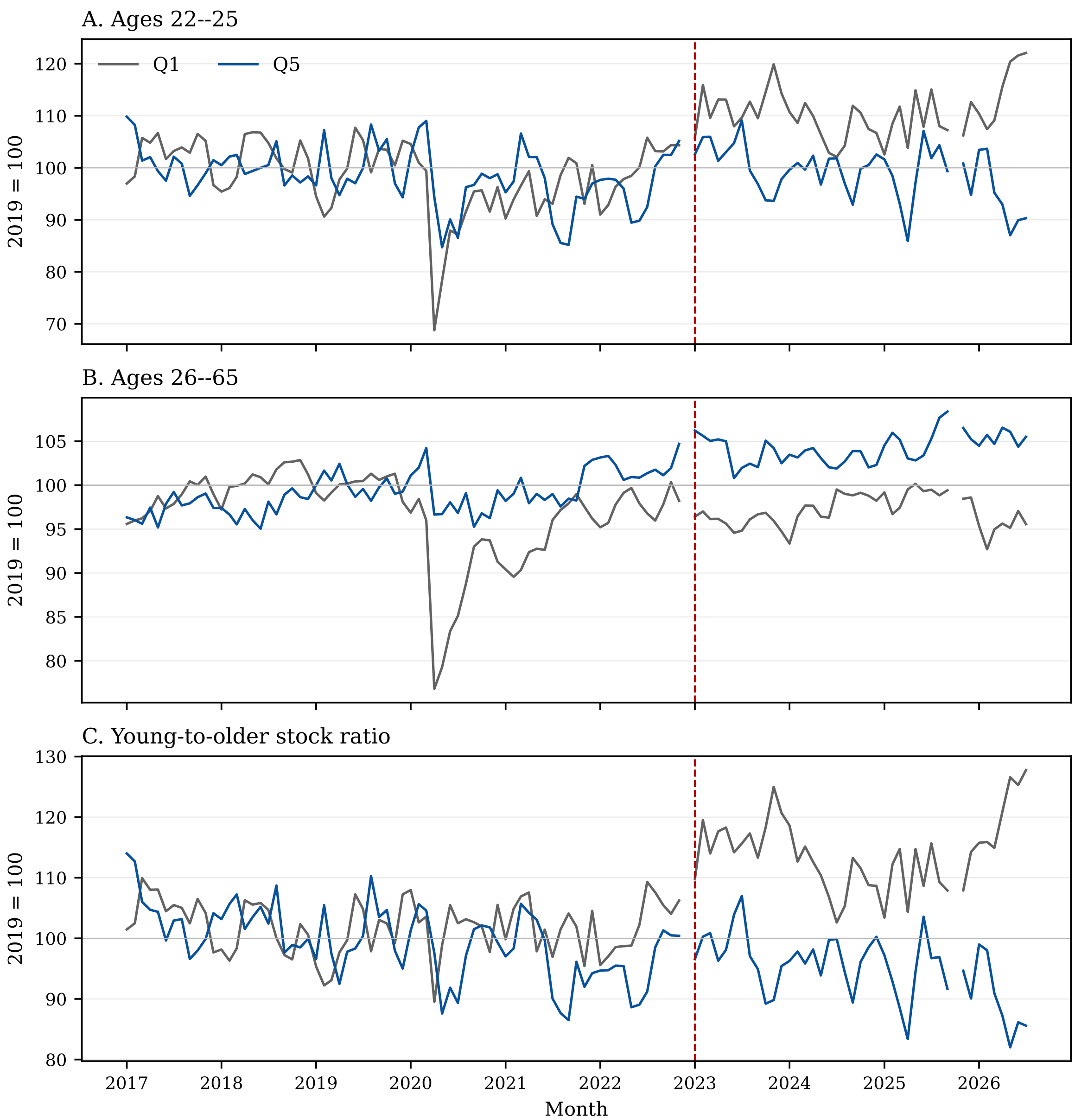

FIGURE 1. Young and older employment stocks in the exposure tails

Notes: Monthly CPS-weighted stocks use rebuilt beta labels and are indexed to each series' 2019 mean. The ratio panel divides young by older stock before indexing. The vertical line marks January 2023; December 2022 is the transition month, and October 2025 is absent rather than interpolated. Paths are descriptive.

The leading computerization measure is the May 2020 O*NET rating "Interacting With Computers." On common support the beta-plus-Webb coefficient is -0.107. Adding computer use makes it -0.212; the paired movement is -0.105 ($[-0.183, -0.027]$). Computer use acts as a suppressor, rejecting the simple claim that this control necessarily attenuates the CPS contrast but not isolating AI. Remotability alone gives -0.116 and an undetected -0.009 movement. Rebuilt-treatment checks of wages, education requirements, routine/manual content, and pandemic shortfalls likewise do not yield individually detected movements on their declared maximal supports. The appendix separately reports the 341-occupation cumulative common-support sequence. These are exploratory nondetections, not equivalence results.

TABLE 3. Exposure conditioning on selected occupational dimensions

Conditioning dimension	Occupations	Baseline	Augmented	Paired change [95% CI]
O*NET computer use	408	-0.107	-0.212	-0.105 $[-0.183, -0.027]$
Dingel–Neiman remotability	408	-0.107	-0.116	-0.009 $[-0.044, 0.027]$
Broad industry in microcells	468	-0.138	-0.099	0.039 $[0.008, 0.070]$

Notes: The first two rows use the same 408-occupation support and reconstructed treatment; each characteristic is interacted with young × post. The industry row changes the cell objective to occupation × broad-industry × age × month and gives 13 industries distinct young-relative post changes. Common bootstrap draws form paired intervals. Coefficient survival or movement is descriptive and does not identify a causal mechanism.

Broad-family paths and a computer-use control need not move the coefficient alike: the former remove all family-common evolution, while the latter partials one correlated dimension. Neither is a causal technology decomposition.

Industry conditioning is built from occupation–industry–age–month cells rather than dominant-industry labels. Allowing 13 broad industries their own young-relative post shifts moves the stratified coefficient from -0.138 to -0.099; the paired 0.039 interval $[0.008, 0.070]$ excludes zero. It does not measure firm AI adoption. BA and non-BA estimates are -0.138 and -0.123; their paired difference is -0.015 ($[-0.156, 0.125]$), with $\mathrm{MDE}_{80} = 0.201$. Exact-age estimates vary but simultaneous paired intervals do not isolate a particular age mechanism.

The industry row changes the cell objective as well as the nuisance set, so it is reported separately from occupation-level characteristic controls. Education and exact-age estimates use literal common supports and common draws; no subgroup is selected by its point estimate. The age coefficients are jointly unequal, but each simultaneous age-minus-pooled interval contains zero.

Table 4 separates uncertainty targets. Occupation wild scores allow serial dependence within occupation. A 22-family sensitivity permits common broad-family shocks. Household full-refit multipliers preserve repeated observations and co-residence under released weights but are not full CPS design inference. None covers exposure labels or route allocation.

TABLE 4. Inference sensitivities for the core occupational comparison

Procedure and target	Pooled 95% CI	Conditioned 95% CI	Paired-movement 95% CI
Same pooled/SOC2-by-calendar-month targets			
Occupation wild score, SOC2-by-month	$[-0.221, -0.044]$	$[-0.161, 0.117]$	$[0.011, 0.210]$
Twenty-two-family Webb wild score	$[-0.211, -0.053]$	$[-0.153, 0.110]$	$[0.006, 0.215]$
Sampling-oriented companion with SOC2-by-post target			
Household full refit, SOC2-by-post	$[-0.179, -0.080]$	$[-0.112, 0.056]$	$[0.037, 0.172]$

Notes: The first two rows estimate the same reconstructed pooled and SOC2-by-calendar-month targets. The 22-family row permits common shocks within broad occupational families. The 199-draw household full-refit sensitivity is conditional on released CPS final weights and uses the less saturated family-by-post companion, so it is separated rather than presented as inference for the same conditioned parameter. The procedures target different perturbations and are not mechanically combined.

With 22 family clusters, the paired movement remains detected under Rademacher and Webb weights; the Webb interval is $[0.0059, 0.2150]$. A calibrated adverse simulation nevertheless rejects a zero pooled effect in 26.7 percent of null replications and the family-conditioned effect in 11.3 percent. This is a stress design, not the CPS sampling law. In the rebuilt-treatment audit, elapsed-calendar HAC leaves scalar uncertainty similar and none of the 15 five-parameter covariance matrices across three targets and five lags is indefinite. Earlier indefinite matrices belong to the superseded historical-treatment audit and remain labeled in the archive.

7 Dynamics, Flows, and External Comparison

7.1 Dynamics

The binary post coefficient is not an event-study design. I estimate quarterly coefficients for all four nonreference quintiles, omitting 2022Q4 (October–November; December is the transition month). The Q5 path has 23 preperiod and 15 postperiod coefficients in pooled and family-month specifications.

Figure 2 shows the rebuilt Q5 path. Joint tests reject that all 23 preperiod coefficients equal zero in both the pooled ($p = .0149$) and family-conditioned ($p = .0053$) models. An insignificant anchored linear slope cannot override those unrestricted tests. January 2023 therefore fails the parallel-evolution diagnostic for causal timing.

The calendar-weighted post functional is distinct from the nonlinear static coefficient. It is

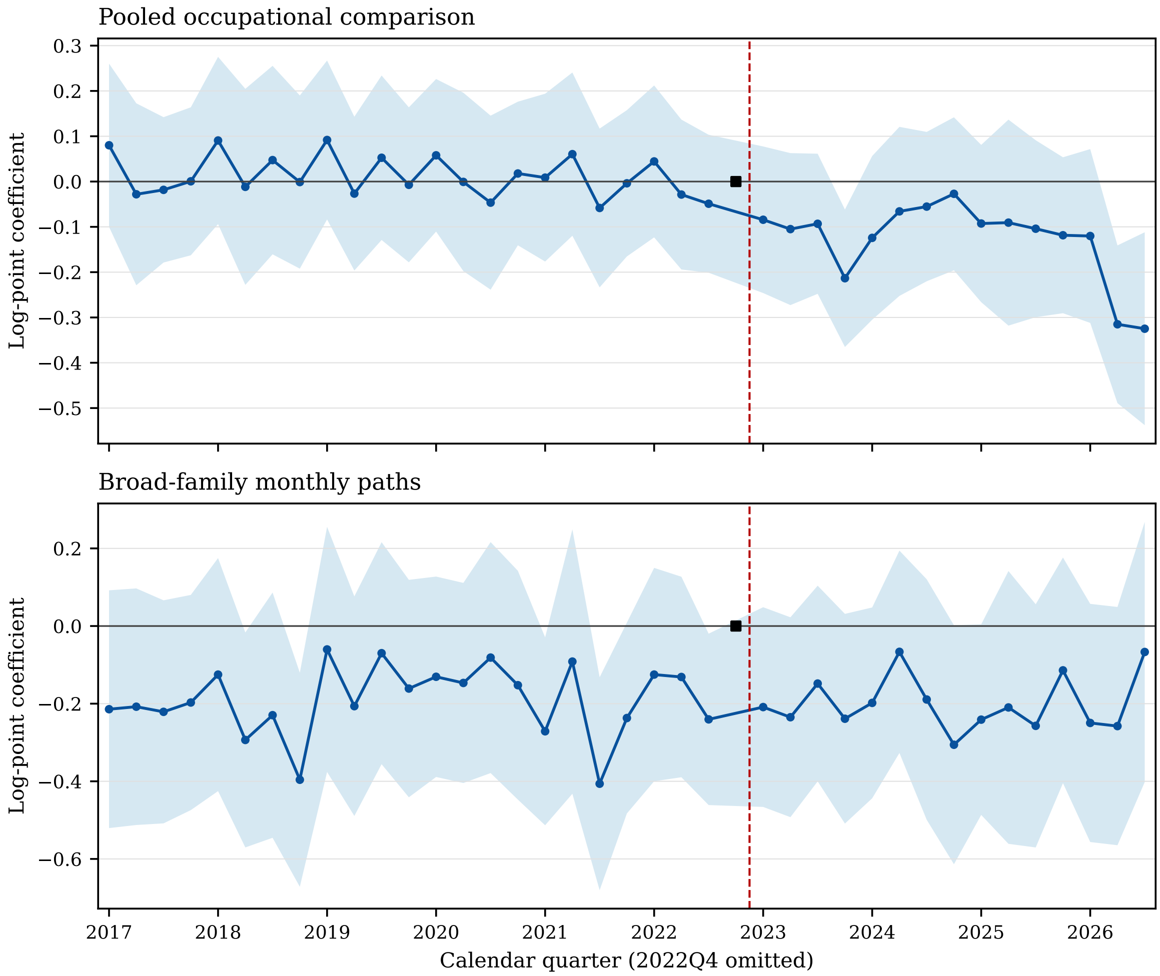

FIGURE 2. Quarterly Q5–Q1 young-relative employment coefficients

Notes: Coefficients are relative to Q1 and omitted 2022Q4 under rebuilt labels; lines are pointwise 95-percent occupation wild-score intervals. The online appendix reports simultaneous bands and the Q2–Q5 system. The conditioned model gives each SOC2 family its own young-relative monthly path.

-0.120 ($[-0.262, 0.022]$) when pooled and -0.207 ($[-0.422, 0.007]$) after family conditioning. Official `HonestDiD` 0.2.8 applies smoothness and relative-magnitude restrictions to the full event vector and covariance. Both conventional intervals already include zero, so no positive zero-exclusion threshold exists. The exercise cannot repair rejected preperiod equality or identify adoption.

Onset estimates from November 2022 through June 2023 are similar within each structure, but common timing does not distinguish AI from coincident shocks. Ending in December 2024 gives a pooled coefficient of -0.111 ($[-0.201, -0.021]$); its paired movement relative to the July-2026 estimate is 0.021 ($[-0.007, 0.049]$), so the design does not detect a change from extending the endpoint beyond 2024. Ending in late 2025 also gives a pooled coefficient near -0.111, versus -0.132 through July 2026, but those paired endpoint movements are detected. Splitting the postperiod instead gives -0.111 in 2023–24 and -0.166 in 2025–26; their paired difference is -0.056 ($[-0.123, 0.012]$), so the official-weight design does not detect a change between eras. Family-conditioned endpoint changes remain near zero. Survey-gap exclusions are small. Declared post-2020 and saturated seasonal models that did not converge remain recorded failures.

7.2 Linked CPS margins

Table 5 estimates distinct adjacent- and twelve-month margins using validated links and official longitudinal weights. Every primary interval includes zero. Normal-theory MDE_{80} values range from 0.115 to 0.336 log point, so these are low-resolution nondetections rather than evidence of absent flow responses.

TABLE 5. Corrected CPS flow margins

Endpoint and margin	Raw link records at risk / entries	Coefficient	95% interval	Normal MDE_{80}
Adjacent employment exit	3,346,227	0.132	$[-0.042, 0.307]$	0.254
Adjacent reported-occupation change	3,207,598	0.003	$[-0.118, 0.123]$	0.174
Adjacent entry-destination allocation	96,981	-0.070	$[-0.250, 0.110]$	0.260
Twelve-month employment exit	1,399,376	0.123	$[-0.111, 0.357]$	0.336
Twelve-month reported-occupation endpoint change	1,110,024	-0.018	$[-0.097, 0.061]$	0.115
Twelve-month entry-destination allocation	82,573	-0.052	$[-0.251, 0.147]$	0.282

Notes: Counts are repeated endpoint-link records, not unique persons; an individual may contribute more than one eligible link. Coefficients are post-January-2023 changes in the ages-22–25 versus ages-26–65, beta-Q5 versus beta-Q1 log rate ratio. Entry destination is a log allocation ratio conditional on an observed entry, not an employment-finding or employer-hiring rate. Primary intervals use 9,999 occupation wild-score draws and official longitudinal weights. $\text{MDE}_{80} = 2.80\widehat{SE}$ uses the analytic occupation-cluster standard error and is a normal-theory precision diagnostic, not the wild-score rejection threshold and not evidence that a smaller effect is absent.

Employment exit starts from employed origins; reported occupation change requires employment at both endpoints; entry destination conditions on nonemployment-to-employment links and measures allocation across exposure groups. The last is neither a finding probability nor employer hiring. Twelve-month links are endpoints, not sums of monthly transitions, and retain 50.98 percent of eligible young origins versus 70.55 percent of older origins. Hours (-0.119, $[-0.295, 0.057]$) and conditional weekly earnings (-0.0245, $[-0.1024, 0.0534]$) are also imprecise and condition on employment; the earnings series ends in March 2023 and contains only three postperiod months.

All flow exposure is defined at the origin where an origin occupation exists. No current occupation is imputed to a nonemployed respondent. Annual endpoints miss intervening jobs and transitions, and positive longitudinal-weight eligibility changes the population; neither horizon is a mechanical decomposition of the stock coefficient.

7.3 Payroll bridge and exposure architecture

BCC's August 2026 19-percent "kept pace" shortfall is a stock-growth comparison, not this regression coefficient or a hiring rate. Complete public membership and tie rules are unavailable, so I use an equal-occupation, tie-preserving approximation and label it a bridge. Its top-two-versus-bottom-three CPS coefficient is -0.0720 ($[-0.1218, -0.0222]$); family-month conditioning gives -0.0168 ($[-0.0810, 0.0474]$), with an undetected paired movement. Differences in universe, ages, outcome, endpoints, employer controls, and worker–firm flows remain.

The bridge therefore asks whether a closer public grouping changes the CPS stock pattern; it does not calibrate CPS transitions to proprietary employer flows. The required entry, exit, job-to-job, occupation-switching, aging, and denominator quantities are not jointly observed.

Finally, $D+\lambda S$ varies the weight on tasks requiring complementary software. Each row uses rebuilt weights and common draws, and $\lambda = .5$ exactly reproduces beta. Native-tail coefficients range from -0.103 at $\lambda = 0$ to -0.156 at $\lambda = 1$; every comparison with beta includes zero, even though 61.6 percent of stock changes quintile at $\lambda = 0$. This is large classification movement with limited coefficient resolution, not evidence about deployed technology.

8 Conclusion

Public CPS data contain a sizeable negative association between GPT-4 task exposure and employment for ages 22–25 relative to ages 26–65. The rebuilt Q5–Q1 coefficient is -0.132 log points, but it falls to an imprecise -0.022 when broad occupation families receive their own young-relative monthly paths. Their paired movement is detected; it is a change between

TABLE 6. Weighting direct and software-complement tasks

λ in $D + \lambda S$	Q5–Q1 coefficient	Difference from $\lambda = .5$ [95% CI]	Stock changing group (%)
0.00	-0.103	0.029 $[-0.044, 0.103]$	61.6
0.25	-0.101	0.031 $[-0.005, 0.068]$	25.5
0.50	-0.132	—	0.0
0.75	-0.143	-0.011 $[-0.039, 0.017]$	10.2
1.00	-0.156	-0.024 $[-0.064, 0.017]$	25.5

Notes: All rows use 468 occupations, reconstructed preperiod weights, tie-preserving native groups, Webb software conditioning, and 9,999 common occupation draws. "Stock changing group" is preperiod employment in occupations whose quintile differs from beta ($\lambda = .5$). The coefficient path is descriptive of the capability boundary; no paired beta comparison detects a difference at five percent.

conditional estimands, not a causal composition share.

Direct within-family tails cover only 29 occupations and are highly imprecise. Preperiod computer use strengthens rather than attenuates the coefficient, showing that computerization is not settled by one control. Unrestricted dynamics reject joint preperiod equality, and linked-CPS flow estimates do not identify a mechanism or replicate employer-match hiring and separation.

The defensible conclusion is therefore comparison-specific. Mapping, support, representation, and conditioning determine what an exposure coefficient measures. A detected movement does not identify why it occurs; nondetection does not establish equivalence. Causal work requires observed adoption or credible assignment and timing variation. This chapter instead establishes that the public-CPS young-worker pattern is principally a broad occupational comparison and quantifies what direct within-family and flow evidence cannot resolve.

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